Registered Proper-Arc Chains and Placement Fibers in a Circular Collapse System

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Abstract

Consider identity-placed singleton packets on an n-cycle, acted on by rotation p and the adjacent idempotent $d : 1 \mapsto 0$. Register an actual fusion that creates a prescribed proper source arc F of length r, and retain its ordered parent-to-port origin, every later fusion involving its carrier, and the complete terminal placed partition. For $n \geq 3$ and $2 \leq r < n$, we classify these generated summaries by a dependent bijection: an internal origin cut, a ternary code for the selected absorption chain, and an interval-cut and occupied-site parameter for the remaining packets. Every parameter is realized by one continuous raw p, d execution. If the chain leaves k atoms outside its carrier, its fixed-origin placement fiber has size $n\binom{n+k-2}{k}$. Summing the fibers gives an exact formula for the complete registered-summary set of every proper source arc. At $r = 2$ and $F = \{0, 1\}$ this recovers the fixed-pair classification. For each fixed r, the normalized diagonal coefficients and the unscaled count have algebraic generating functions, and the count is asymptotic to $\frac{4(r-1)}{3^{r+2}} n(9/2)^n$.

Keywords: circular packet dynamics, registered fusion, source-addressed history, absorption chain, dependent enumeration

1 Introduction

A final placed partition does not say which fusion created a selected packet or which later fusions involved it. For the circular action studied here, we count the complete summaries obtained by recording that origin, the selected carrier's ordered absorption chain, and the terminal placement. The selected packet is any proper source arc, not necessarily a pair. Its two registration parents may have different sizes, while unselected fusions can assemble multi-atom partners and change the packets left outside the carrier. The case $F = \{0, 1\}$ was classified in the fixed-pair HAL preprint (Chen, 2026); Section 7 identifies its exact specialization here.

The main statement classifies *generated* summaries. Write $m = n - r$. An origin has an internal cut $t \in \{1, \dots, r-1\}$; a selected chain has a code $v \in \{0, L, R\}^m$; and its terminal placement belongs to a fiber $\mathsf{Fib}_{n,m-b(v)}$, where $b(v)$ is the last nonzero checkpoint or zero. We prove

$$S_F \simeq \coprod_{t=1}^{r-1} \coprod_{v \in \{0,L,R\}^m} \mathsf{Fib}_{n,m-b(v)}. \tag{1.1}$$

The origin index is already visible in the smallest case beyond a pair. Take $n = 4$ and $F = \{0, 1, 2\}$. The chronological words p^3dpd and dp^3dpd register F with respective origins

$$(d, (0, \{0\}), (1, \{1, 2\}), 0) \quad \text{and} \quad (d, (0, \{0, 1\}), (1, \{2\}), 0).$$

Both end with F at site 0, $\{3\}$ at site 3, holes at sites $1, 2$, and an empty selected chain. The placed partition alone cannot recover which internal cut registered F.

The proof constructs one execution from each parameter tuple: hole slides route packets without fusion, checkpoint words assemble and absorb the chosen partner, and outside fusions followed by placement control reach the specified terminal state. The inverse reads the cut, code, and placement parameters from the complete summary. Sections 2–5 establish the classification, Section 6 counts its fibers, and Sections 7–8 derive the fixed-pair specialization, generating functions, and asymptotics.

The algebraic setting is adjacent to McAlister's study of semigroups generated by a group and an idempotent, including its cyclic-group case (McAlister, 1998), and Mitchell–Levi's work on generating ranks for endomorphisms of finite circular orders (Mitchell and Levi, 2006). On the combinatorial side, Mansour–Munagi enumerate set partitions by circular successions (Mansour and Munagi, 2014). Those works study transformation-semigroup structure or partition statistics. The principal contribution here is the bijection between complete generated summaries and a dependent family of origin, chain, and placement parameters, with realizability established by explicit executions.

2 Action, Records, and Placement Control

Let $n \geq 3$. The source atoms Ω and physical sites Q are distinct sets, each identified with $\mathbb{Z}/n\mathbb{Z}$ at the singleton seed. A placed partition P assigns a nonempty, disjoint atom block to each occupied site; its blocks cover Ω. Unoccupied sites are holes. The seed is $P_\varnothing(q) = \{q\}$. The letters act on sites by

$$p(q) = q + 1, \qquad d(1) = 0, \qquad d(q) = q \quad (q \neq 1). \tag{2.1}$$

Their action on placements is pushforward union:

$$(a_\# P)(y) = \bigcup_{q:\, a(q)=y} P(q), \qquad a \in \{p, d\}, \tag{2.2}$$

with empty values omitted. Words are executed chronologically, left to right. A fusion occurs precisely when d sees occupied ports 0 and 1. Its full record is

$$\rho = (d, (0, U), (1, V), 0, U \cup V), \tag{2.3}$$

where U and V are the actual parent blocks at those ports. In particular, their port binding is part of the record. The registration origin is the four-component projection of this five-component fusion record:

$$\mathrm{Orig}(\rho) = (d, (0, U), (1, V), 0). \tag{2.3a}$$

For $c \in \Omega$ and $1 \leq \ell \leq n$, put $U_{c,\ell} = \{c, c+1, \ldots, c+\ell-1\}$, with source labels modulo n. Fix a proper arc

$$F = U_{c,r}, \qquad 2 \leq r < n, \qquad m = n - r, \tag{2.4}$$

and write its positive source complement as the linear list

$$B = (c + r, c + r + 1, \ldots, c + n - 1). \tag{2.5}$$

An execution registers F only when one of its actual fusions first creates the block F. The origin o_F is $\mathrm{Orig}(\rho)$ for that fusion; on registered histories its omitted result is the fixed block F. Thus equality of origins is equivalent to equality of the complete registration records within S_F. After registration, the unique packet containing F is the *carrier*. Let $A_F(h)$ be the chronological list of all later fusion records (2.3) in which this carrier is a parent; fusions wholly outside it are not selected. For a registered history h set

$$\Gamma(h, F) = (P_h, F, o_F, A_F(h)), \qquad S_F = \{\Gamma(h, F) : h \text{ is registered for } F\}. \tag{2.6}$$

Equality in S_F is equality of all four fields. Execution words and unselected fusion records are not fields of a summary.

Lemma 2.1 (source order). At every generated placement, the packets are disjoint nonempty source arcs, and their positive physical cyclic order is their positive source cyclic order, up to rotation. This includes an arc wrapping across the source label $n-1 \to 0$.

Proof. The seed has the asserted order. A p step rotates occupied sites without changing blocks. If d sees both ports empty or only port 0 occupied, nothing changes. If only port 1 is occupied, its packet moves to the empty predecessor 0 and passes no occupied site. If both ports are occupied, those packets are consecutive in physical cyclic order and hence in source cyclic order when at least three packets remain. With exactly two packets their union is all of Ω, also a source arc. Replacing the two adjacent packets by their union preserves order. These cases exhaust each letter, including the physical boundary $n-1 \to 0$. □

For a site j, write $[j]_n \in \{0, \dots, n-1\}$ for its nonnegative representative. The chronological raw word

$$E_j = p^{[-j]_n}\, d\, p^{[j]_n} \tag{2.7}$$

sends a packet at $j+1$ to j at its endpoints and fixes all other sites. If j is empty, its middle d sees an empty port 0 and causes no fusion. The rotations are also nonfusing. Thus E_j is an actual hole slide, not an additional generator. During the prefix and suffix rotations the current placement is still obtained from the preceding concrete placement by the indicated letter; occupancy of both ports during a rotation does not create a fusion.

Lemma 2.2 (placement control). Two placements of the same named packets in the same positive cyclic packet order are joined by a fusion-free raw p, d word. The word acts on the current concrete placement at each step.

Proof. With one packet, a rotation suffices even if there are holes. Suppose there are $s \geq 2$ packets $B_0, \dots, B_{s-1}$ in their common cyclic order. Let h_i count holes after B_i and before B_{i+1}. If $h_i > 0$, apply (2.7) at the hole just before B_{i+1}. It changes only

$$(h_i, h_{i+1}) \longmapsto (h_i - 1, h_{i+1} + 1), \qquad (s \geq 2,\ h_i > 0). \tag{2.8}$$

Given target gaps h_i^*, choose an excess gap and a deficit gap and route one hole forward along the directed gap cycle. The first donor has a hole; each later donor has just received it. Intermediate gaps have zero net change, so $\sum_i (h_i - h_i^*)_+$ decreases by one. Repetition reaches the target gaps. A final power of p aligns B_0 and consequently all packets. If no holes exist, this last rotation already suffices. Every slide was nonfusing on its actual predecessor placement. □

Corollary 2.3 (ordered-port admission). Two packets consecutive in positive source order can be placed at ports $(0, 1)$ in that order, with all other packets elsewhere, and fused by one d.

Proof. Choose such a target placement in the same cyclic packet order and apply Lemma 2.2. The next d is its sole fusion. □

Corollary 2.4 (two-packet port exception). If exactly two named packets X, Y remain, both (X, Y) and (Y, X) can be placed at ports $(0, 1)$ by fusion-free words. If at least three packets remain and X, Y are consecutive in positive source order, only (X, Y) is compatible with those ordered ports while the named packets remain separate.

Proof. A two-packet cyclic order has both (X, Y) and (Y, X) as rotations, so Lemma 2.2 admits both target placements. With at least three packets, reversing the ports would make Y immediately precede X in the physical positive order, contrary to the source order in Lemma 2.1. □

3 Registration Origins and a Common Boundary State

For $1 \le t < r$ define the registration origin

$$\rho_t = \big(d, (0, U_{c,t}), (1, U_{c+t,r-t}), 0, F\big), \qquad o_t = \mathrm{Orig}(\rho_t). \tag{3.1}$$

Proposition 3.1 (origin classification). The origins of F are exactly $o_1, \dots, o_{r-1}$. Each is realized from the singleton seed by an actual word ending at the same post-registration placement

$$P^{\mathrm{bdry}}(0) = F, \quad P^{\mathrm{bdry}}(i) = \varnothing \ (1 \le i < r), \quad P^{\mathrm{bdry}}(i) = \{c + i\} \ (r \le i < n). \tag{3.2}$$

Their post-registration summaries differ only in the recorded origin.

Proof. Immediately before a fusion creating F, its two parent packets partition F and are source arcs by Lemma 2.1. Their only possible split is the internal cut t, giving the two sets in (3.1). Since F is proper, at least one outside packet is present. The source cycle therefore reads prefix parent, suffix parent, outside packet(s). By Corollary 2.4, the prefix must be at port 0 and the suffix at port 1: reversing them would put an outside packet between those parents in the directed order.

For realization, first apply $p^{[-c]_n}$ to the seed. In this aligned frame atom $c + i$ sits at physical site i. Read the following E-words in decreasing index order, omitting empty ranges:

$$\underbrace{E_{t-2}E_{t-3}\cdots E_0}_{\text{left parent}} \quad \underbrace{E_{r-2}E_{r-3}\cdots E_t}_{\text{right parent}} \quad \underbrace{E_{t-1}E_{t-2}\cdots E_1}_{\text{right slide}} \quad d. \tag{3.3}$$

In the left loop, immediately before E_j, the singleton $\{c + j\}$ occupies site j, the already assembled interval $U_{c+j+1,t-j-1}$ occupies $j + 1$, and $j + 2, \dots, t - 1$ are holes. The middle d therefore fuses precisely those packets; after E_j, $U_{c+j,t-j}$ occupies j and $j+1$ is empty. Descending to $j = 0$ leaves $U_{c,t}$ at site 0. The right loop has the same induction on sites $t, \dots, r - 1$, disjoint from the left parent; it leaves $U_{c+t,r-t}$ at site t. Sites $1, \dots, t - 1$ are holes. In the last loop, immediately before E_j, the right parent is at $j+1$ and j is empty. Each step slides it left without fusion, ending at site 1. The final d creates F with the stated port binding. All earlier fusions occurred inside one strict parent subarc, so this is the first creation of F. Outside atoms stayed singleton at sites $r, \dots, n - 1$; the resulting placement is (3.2), independent of t. The cases $t = 1$ and $t = r - 1$ simply make the corresponding loops empty. □

Call an actual registered state a **clean boundary state** if its carrier C is a packet containing F, its complement in positive source order is a list of singleton packets B (possibly empty), and its origin and selected chain are fixed as (o, A). Proposition 3.1 supplies one with $C = F$ and $A = ()$ for each origin.

4 Ternary Selected Chains

Put $\mathcal{V}_m = \{0, L, R\}^{\{1,\dots,m\}}$. For $v \in \mathcal{V}_m$, let $0 < b_1 < \cdots < b_j \le m$ be its nonzero positions and set $b_0 = 0$. Starting with carrier F and complement list (2.5), at checkpoint b_i take $\delta_i = b_i - b_{i-1}$ atoms. If $v(b_i) = R$, take the first δ_i atoms of the current complement and put (C, D) at ports $(0, 1)$; if $v(b_i) = L$, take its last δ_i atoms and put (D, C) there. Record (2.3) and replace C by $C \cup D$. Let $D_F(v)$ be the resulting ordered record list, and define $b(v) = b_j$ or 0 if $j = 0$. The

decoder is independent of the origin cut t: Proposition 3.1 gives every o_t the same post-registration placement, while the subsequent record construction reads only F and v.

Lemma 4.1 (actual checkpoint extension). From any clean boundary state with nonempty singleton complement B, any prescribed nonempty prefix or suffix D of B can be assembled outside the carrier and absorbed in one selected fusion with the specified R or L port orientation. The resulting actual state is clean, its origin is unchanged, and its selected chain has exactly the prescribed one record appended.

Proof. List the atoms of D in positive source order. Merge its first two singleton packets, then that outside packet with the next singleton, and so on. Before each merge the two named packets are source neighbors; Corollary 2.3 places them at ordered ports along a fusion-free word. The following d merges only those outside packets. Inductively the partial packet is the first h atoms of D, every other outside atom remains singleton, and neither the origin nor the selected chain changes. There are $|D| - 1$ outside fusions, or none when $|D| = 1$.

Now D is one packet adjacent to the carrier in the source packet cycle: it follows C for a prefix and precedes C for a suffix. Use Corollary 2.3 again to put (C, D) or (D, C) at ports $(0, 1)$. The next d creates exactly the ordered record declared by the code. Its new carrier is $C \cup D$; removing D leaves a singleton complement interval. Thus the resulting state is a clean boundary state with the origin unchanged and exactly one new selected record. If $D = B$, that complement is empty and no further checkpoint is required. If only two packets remain, both ordered port choices are admitted by Corollary 2.4. The routing words act successively on these placements, yielding one continuous execution. □

Theorem 4.2 (fixed-origin chain bijection). For each origin o_t, D_F maps $\mathcal{V}_m$ bijectively onto the attainable selected chains with that origin. There are 3^m chains; exactly $\binom{m}{j}2^j$ have j selected absorptions.

Proof. Lemma 4.1, iterated from Proposition 3.1's actual boundary state, realizes every code and ends after each checkpoint with the clean condition needed by the next. Conversely, consider any registered history. Its carrier is a source arc containing F, so its complement is a linear source interval. Lemma 2.1 forces the partner in a selected fusion to be a nonempty prefix or suffix of that interval, although outside-only fusions may have assembled it earlier. Its full record identifies the carrier parent and its port, hence R or L. The cumulative absorbed sizes $|C_i| - r$ strictly increase in $\{1, \dots, m\}$ and recover the nonzero positions of v; all other positions are zero. At the final checkpoint, even if the carrier becomes Ω, the two possible port bindings remain distinct records. This encoding is inverse to the construction. Choosing j nonzero positions and their signs gives the refined count. □

5 Complete Placement Fibers and the Dependent Bijection

Fix an origin o_t and code v. Let C_v be its final carrier and put

$$k = |\Omega \setminus C_v| = m - b(v). \tag{5.1}$$

When $k > 0$, the outside atoms form a linear source interval. Any interval partition of this complement may be specified by a cut set $K \subseteq \{1, \dots, k-1\}$. If it has $u = |K| + 1$ outside blocks, choose $u + 1$ occupied sites $S \subseteq Q$ and distinguish one site $q_C \in S$ for the carrier. Assign the outside blocks to the other sites in positive cyclic order following q_C. Define

$$\begin{aligned} &\mathsf{Fib}_{n,k} = \{(K, S, q_C) : K \subseteq \{1, \dots, k-1\}, |S| = |K| + 2,\ q_C \in S\}, \quad k > 0, \\ &\mathsf{Fib}_{n,0} = Q. \end{aligned} \tag{5.2}$$

For $k = 0$, the parameter is the sole carrier site.

Lemma 5.1 (literal placement uniqueness). Fix a carrier source arc C and decode $\pi \in \mathsf{Fib}_{n,k}$ into the placed partition specified by (5.2). If two valid parameters π, π' decode to the same literal placement, then $\pi = \pi'$.

Proof. When $k > 0$, the placement determines its support S, the unique site q_C occupied by C, and the internal source cuts K between its outside packets. These are exactly the three components of π. When $k = 0$, the sole packet is $C = \Omega$ and its site is the parameter. Thus equality of placements gives equality of parameters in both cases. □

Theorem 5.2 (fixed-chain placement fiber). The summaries with fixed $(F, o_t, D_F(v))$ are in bijection with $\mathsf{Fib}_{n,k}$. They are exactly the placements with packet C_v, an interval partition of its source complement, and the same positive cyclic packet order. In particular,

$$|\mathsf{Fib}_{n,k}| = n\binom{n+k-2}{k}, \tag{5.3}$$

and, for $k > 0$, the rank-s part has size

$$s\binom{k-1}{s-2}\binom{n}{s}, \qquad 2 \le s \le k+1. \tag{5.4}$$

Proof. In any actual summary with the fixed chain, the unique packet containing F is C_v: each selected record determines its next carrier, and a later fusion involving the carrier would add another selected record. Every outside packet is an interval in the linear complement by Lemma 2.1. The terminal placement therefore determines its outside cuts, occupied sites, and carrier site uniquely, giving (5.2).

For the converse, start from the clean endpoint constructed in Theorem 4.2. Fix the target cuts K. While some target outside interval contains two current packets, choose neighboring packets within that interval. Corollary 2.3 routes them to ports $(0, 1)$ without fusion; one d merges them. Neither parent contains F, so the origin and selected chain remain fixed. If N_T is the number of current packets within target interval T, the integer

$$\sum_{T \text{ target outside interval}} (N_T - 1) \tag{5.5}$$

decreases by exactly one at every such merge. It reaches zero precisely when every target outside interval has become one packet. The carrier and outside blocks now have the target positive source order. Lemma 2.2 moves this actual placement to the prescribed occupied sites without fusion. For $k = 0$, rotate the sole carrier to the prescribed site. Thus every parameter in (5.2) gives a generated summary, and the preceding readout is its inverse by Lemma 5.1.

If $k > 0$ and there are u outside blocks, there are $\binom{k-1}{u-1}$ cuts. There are $\binom{n}{u+1}$ choices of occupied sites and $u + 1$ choices of the carrier site. This gives (5.4) with $s = u + 1$. Summing and using Vandermonde's identity,

$$\begin{aligned}\sum_{u=1}^{k}(u+1)\binom{k-1}{u-1}\binom{n}{u+1} &= n\sum_{u=1}^{k}\binom{k-1}{u-1}\binom{n-1}{u} \\ &= n\binom{n+k-2}{k}.\end{aligned} \tag{5.6}$$

For $k = 0$ there are n carrier sites, agreeing with (5.3). □

Define the dependent parameter set

$$\mathcal{D}_{n,r} = \coprod_{t=1}^{r-1} \coprod_{v \in \mathcal{V}_m} \mathsf{Fib}_{n,m-b(v)}. \tag{5.7}$$

Theorem 5.3 (complete generated-summary classification). For $n \geq 3$ and $F = U_{c,r}$ with $2 \leq r < n$, there is a bijection

$$S_F \simeq \mathcal{D}_{n,r}. \tag{5.8}$$

It retains the actual origin, full ordered selected chain, and terminal placed partition together. Every parameter tuple in $\mathcal{D}_{n,r}$ is realized by one continuous chronological p, d word from the singleton seed.

Proof. Given (t, v, π), define Decode(t, v, π) as the complete summary with origin o_t, selected chain $D_F(v)$, and terminal placement specified by π. Proposition 3.1, Lemma 4.1 and Theorem 5.2 concatenate to realize these three fields along one raw history, so this tuple belongs to S_F.

Conversely, for $\Gamma = (P, F, o, A) \in S_F$, Proposition 3.1 reads the unique cut t from o. Theorem 4.2 reads the unique code v from the full ordered record list A. The carrier is the packet of P containing F, and is the carrier C_v decoded from those records. Theorem 5.2 reads the unique outside cut set, occupied-site set and carrier site π, or just that site when $k = 0$. Put Encode$(\Gamma) = (t, v, \pi)$.

Starting with parameters, encoding their decoded summary recovers t from its ordered origin, v from its full selected records, and π from its terminal packets and sites by Lemma 5.1. Thus Encode $\circ$ Decode $=$ id. Starting with a generated summary, the decoded origin and chain equal its original fields. Its recovered cuts and sites determine the same outside packets at the same coordinates in positive source order, so the terminal placement agrees site by site. Hence Decode $\circ$ Encode $=$ id. These equalities concern summaries. □

6 Exact Enumeration

Let $s_{n,r} = |S_F|$ for any fixed proper source arc of length r. Theorem 5.3 shows that this number is independent of its starting atom c. There is one all-zero code, with $b(v) = 0$. For $1 \leq b \leq m$, exactly $2 \cdot 3^{b-1}$ codes have last nonzero checkpoint b: choose one of L, R at b, choose arbitrary earlier symbols, and put zeros later. Their respective outside sizes are m and $m - b$. Equation (5.3) therefore gives the exact total.

Theorem 6.1 (proper-arc summary count). For $n \geq 3$, $2 \leq r < n$, and $m = n - r$,

$$\boxed{s_{n,r} = (r-1)n \left[\binom{n+m-2}{m} + 2 \sum_{b=1}^{m} 3^{b-1} \binom{n+m-b-2}{m-b} \right].} \tag{6.1}$$

Proof. By Theorem 5.3 and (5.3), sum the fixed-chain fibers over the codes just counted. The $r - 1$ origin components are disjoint because origin is a field of the summary. □

At the smallest complement size $m = 1$ (that is, $r = n - 1$), the empty chain has $n(n-1)$ placements per origin and each of the two one-record chains has n. Thus (6.1) gives $s_{n,n-1} = (n-2)n(n+1)$, including $s_{3,2} = 12$.

For a fixed origin, the chain-length refinement is immediate: exactly $\binom{m}{j} 2^j$ attainable selected chains have j records. If a nonempty chain has final checkpoint b and exactly j records, its multiplicity is $2^j \binom{b-1}{j-1}$ and its fixed-chain fiber has $n\binom{n+m-b-2}{m-b}$ placements. These refine (6.1).

7 Exact Relation to the Fixed-Pair Result

Let S_n^{pair} denote the complete registered-summary set of the fixed-pair HAL preprint (Chen, 2026) for $F_0 = \{0, 1\}$, with its origin projection (2.3a), ordered selected records, and literal terminal placement. The following identifies the summary sets directly.

Proposition 7.1 (fixed-pair specialization). Set $r = 2$, $c = 0$, and $n \geq 3$. Then $S_{F_0} = S_n^{\mathrm{pair}}$ as sets of complete summaries. The chain and fiber components of Theorem 5.3 specialize to the fixed-pair preprint's Theorems 3.1 and 4.2; Theorem 6.1 specializes to its Theorem 5.1. Writing $m = n - 2$,

$$\mathcal{D}_{n,2} \simeq \coprod_{v \in \{0,L,R\}^m} \mathsf{Fib}_{n,m-b(v)}, \tag{7.1}$$

and

$$s_{n,2} = n\left[\binom{2m}{m} + 2\sum_{b=1}^{m} 3^{b-1}\binom{2m-b}{m-b}\right]. \tag{7.2}$$

Proof. The sole internal cut is $t = 1$ and its origin is $o_1 = (d, (0, \{0\}), (1, \{1\}), 0)$, the fixed-pair origin. The initial boundary state in Proposition 3.1 is F_0 at site 0 with all outside atoms singleton, as in the fixed-pair construction. Both chain decoders read nonzero code positions as cumulative absorbed sizes and L/R as the carrier's port; hence their complete ordered records agree. For each code, (5.2) uses the same outside cuts, occupied-site set, and carrier site as the fixed-pair fiber parameter (its D, Z, q). Thus the parameter-to-summary maps agree as literal summary maps, though their chosen raw realization words need not be identical. The origin index disappears from (5.7), giving (7.1). Substituting $r = 2$ into (6.1) gives (7.2), the fixed-pair total. □

For $r > 2$, Proposition 3.1 gives an actual registration word for each of the $r - 1$ internal cuts, and these words reach the same post-registration placement. From each such endpoint, Lemma 4.1 and Theorem 5.2 realize every chain code and every parameter in its dependent placement fiber by one continuous execution. Conversely, the ordered parents in the complete origin recover the cut, even when two summaries share a terminal placement. Thus the factor $r - 1$ in (6.1) is a consequence of the disjoint, actually realized components of Theorem 5.3, not a multiplier assumed from the fixed-pair formula.

8 Generating Functions and Fixed-Arc Asymptotics

Theorem 6.1 first gives a generating function with n held fixed. For $n \geq 2$ and formal $m \geq 0$, define

$$A_n(m) = \binom{n+m-2}{m} + 2\sum_{b=1}^{m} 3^{b-1}\binom{n+m-b-2}{m-b}. \tag{8.1}$$

Only $1 \leq m \leq n-2$ indexes a proper registered arc, for which $s_{n,n-m} = (n-m-1)nA_n(m)$. Values at $m = 0$ or beyond this range are formal coefficients, not additional proper-arc counts.

Proposition 8.1 (fixed-n generating function). As a formal power series,

$$G_n(x) := \sum_{m \geq 0} A_n(m)x^m = \frac{1}{(1-3x)(1-x)^{n-2}}. \tag{8.2}$$

Proof. The first term of (8.1) has generating function $(1-x)^{-(n-1)}$. Reindexing the second term by $m - b$ gives $2x(1-3x)^{-1}(1-x)^{-(n-1)}$. Adding them cancels one factor $1 - x$ in the numerator and yields (8.2). □

To follow a fixed source-arc length while n increases, set $Z_m^{(r)} = A_{m+r}(m)$ for fixed $r \geq 2$. Its $m = 0$ term is the formal coefficient 1; terms with $m \geq 1$ correspond to proper arcs. Let

$$t(z)=\sqrt{1-4z}, \qquad q(z)=\frac{2}{1+t(z)}, \tag{8.3}$$

with the square root expanded at $z=0$.

Proposition 8.2 (fixed-r diagonal generating function). For every fixed $r\geq 2$,

$$\sum_{m\geq 0} Z_m^{(r)} z^m = q(z)^{r-2}\frac{1+t(z)}{t(z)(3t(z)-1)}. \tag{8.4}$$

Proof. The binomial-series identity

$$\sum_{k\geq 0}\binom{2k+c}{k}z^k=\frac{q(z)^c}{t(z)}, \qquad c\geq 0, \tag{8.5}$$

can be checked directly by formal coefficient extraction. Set $u=q-1$. The equation $q=1+zq^2$ gives $z=u/(1+u)^2$, $t=(1-u)/(1+u)$, and $dz=(1-u)(1+u)^{-3}du$. Using $[z^k]f(z)=\operatorname{Res}_{z=0} f(z)z^{-k-1}dz$ yields

$$[z^k]\frac{q^c}{t}=\operatorname{Res}_{u=0}\frac{(1+u)^{2k+c}}{u^{k+1}}\,du=\binom{2k+c}{k}.$$

The first term of (8.1), with $n=m+r$, contributes q^{r-2}/t. In the tail put $j=b-1$ and $k=m-b$. Its binomial factor becomes $\binom{2k+j+r-1}{k}$, so (8.5) and the geometric sum in j give a contribution $2zq^{r-1}/[t(1-3zq)]$. Their sum is

$$\frac{q^{r-2}}{t}\frac{1-zq}{1-3zq}=q^{r-2}\frac{1+t}{t(3t-1)}, \tag{8.6}$$

because $zq=(1-t)/2$. □

The preceding function generates normalized coefficients. The original proper-arc summary counts also have an algebraic generating function.

Corollary 8.3 (unscaled proper-arc count series). Write $H_r(z)=\sum_{m\geq 0} Z_m^{(r)} z^m$ for the function in (8.4). For every fixed $r\geq 2$,

$$\sum_{m\geq 1} s_{m+r,r} z^m=(r-1)\big[rH_r(z)+zH_r'(z)-r\big]. \tag{8.7}$$

This series is algebraic over $\mathbb{Q}(z)$.

Proof. For $m\geq 1$, Theorem 6.1 gives $s_{m+r,r}=(r-1)(m+r)Z_m^{(r)}$. The coefficient of z^m in rH_r+zH_r' is $(r+m)Z_m^{(r)}$. Since $H_r(0)=1$, subtracting r removes precisely the formal $m=0$ term, where the source arc would be full rather than proper. Formula (8.4) places H_r in $\mathbb{Q}(z,\sqrt{1-4z})$; this field is closed under differentiation, so the right side of (8.7) is algebraic. □

Theorem 8.4 (fixed-r asymptotic with an explicit error). For each fixed $r\geq 2$ and $n>r$,

$$0\leq 1-\frac{s_{n,r}}{\frac{4(r-1)}{3^{r+2}}n(9/2)^n}\leq \frac{3^{r+2}}{2^{r+4}}\left(\frac{8}{9}\right)^n. \tag{8.8}$$

Consequently,

$$s_{n,r}=\frac{4(r-1)}{3^{r+2}}n\left(\frac{9}{2}\right)^n\left[1+O_r\left(\left(\frac{8}{9}\right)^n\right)\right], \qquad \lim_{n\to\infty} s_{n,r}^{1/n}=\frac{9}{2}. \tag{8.9}$$

Proof. Expanding (8.2) in the opposite order gives

$$A_n(m) = 3^m \sum_{k=0}^{m} 3^{-k} \binom{n+k-3}{k}. \tag{8.10}$$

The infinite sum is $(3/2)^{n-2}$. For $m = n - r$, its omitted tail is nonnegative and, using $\binom{n+k-3}{k} \le 2^{n+k-3}$, is at most

$$2^{n-3} \sum_{k=n-r+1}^{\infty} (2/3)^k = 3 \cdot 2^{n-3}(2/3)^{n-r+1}. \tag{8.11}$$

Dividing by $(3/2)^{n-2}$ gives the upper bound in (8.8), since

$$(r-1)n3^{n-r}(3/2)^{n-2} = \frac{4(r-1)}{3^{r+2}} n(9/2)^n.$$

The limit in (8.9) follows for fixed r. At $r = 2$, (8.8) reads $0 \le 1 - s_{n,2}/[n(9/2)^{n-2}] \le (8/9)^{n-2}$, recovering the fixed-pair growth bound and determining the sign of its error. □

For fixed r, the source-arc length changes the prefactor $4(r-1)/3^{r+2}$, but not the exponential growth base $9/2$.

9 Computational Checks

Exhaustive generated-state enumeration for $3 \le n \le 6$ agrees with the origin classification, chain encoding, fixed-chain fiber cardinalities, and total counts. Direct evaluation of the registration word (3.3) confirms every proper-arc internal cut through $n = 7$. The code and fixed results are under `experiments/proper_arc_registered_summaries/` in the RIME Lite source snapshot. These finite checks complement the all-parameter constructions and inverse maps proved above.

10 Conclusion

For every proper source arc, the complete generated summaries are classified by an origin cut, a selected-chain code, and a placement fiber determined by the final checkpoint. Explicit chronological words realize every parameter tuple, while the inverse reads the same data from the registered summary.

The classification yields exact fiber and total counts, algebraic generating functions, and fixed-arc growth with exponential base $9/2$. At $r = 2$ it specializes to the fixed-pair classification.

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